

ESTELA KATCHEN

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Education

University of Western Ontario & Ivey Business School

London, ON | 2024–Present

Dual degree: BESC (Mechatronics Systems Engineering) & HBA (Business Administration)

- Dean's Honour List (2025–2026)

Experience

Summer AI Research Assistant

May 2026 – Present

Yang's AiLearn Lab, Prof. Yimin Yang, Western University — London, ON

- Extended a multi-view CLIP-based framework from breast density estimation to three-class cancer diagnosis (normal/benign/malignant), adapting the model, data pipeline, and confidence modules across 7 Python/PyTorch files.
- Implemented Dempster-Shafer evidential fusion across four mammogram views to produce per-prediction confidence scores for clinical interpretability; improved classification accuracy 6% over baseline.
- Extending the certainty-estimation framework to ultrasound imaging alongside graduate researchers, targeting international conference submission (Sept 2026); author research briefs on dataset selection and study design for biweekly lab reviews.

Product Analyst Intern

May – Aug 2025

Wahi – RPS (Brookfield Portfolio Company) — Toronto, ON

- Led competitive benchmarking across 6+ valuation technology platforms and 7 product dimensions, identifying gaps in UX, workflows, onboarding, and client reporting.
- Ran 20+ discovery interviews with clients and stakeholders — appraisers, brokers, app users — and analyzed call transcripts in Dovetail against Datadog usage data to pinpoint workflow friction.
- Presented executive-ready strategy materials, including feature matrices and UX teardown reports, to product and engineering stakeholders; delivered 10+ prioritized recommendations adopted by the product team.

Leadership & Involvement

Sponsorship Ambassador (2026-27) | VP Mentorship (2025-26) | First-Year Rep (2024-25)

Sept 2024 – Present

Western Women in Engineering (WiE)

- Founded and launched the career mentorship program for upper-year students, designing the matching framework and running outreach, pairing, onboarding, and feedback end to end.
- Ran the Big Sister/Little Sister program supporting 600+ first-year students; managed mentor recruitment, communications, event logistics, and participant feedback.
- Securing corporate sponsorships across a 2,000+ member organization; leading partner meetings and negotiating event deliverables.

VP Competitions — Western Engineering Competition (WEC)

May 2026 – Present

- Directing end-to-end planning for WEC 2026: scoping competition formats, designing judging rubrics, and coordinating cross-team logistics across 5+ categories.
- Chairing planning meetings with the executive team, faculty advisors, and industry judges to align on timelines, scoring criteria, and deliverables.

Data Team Developer — 3DWestern

May 2026 – Present

- Building data preprocessing pipelines and benchmarking open-source vision-language models to support ML-based 3D print quality classification.

Projects

Colour-Sorting Robot & Autonomous Retrieval Robot

Feb – Apr 2026

- Designed and built an ESP32/C++ robot to autonomously detect and sort coloured objects, splitting subsystem ownership across a 4-person team and integrating movement, sensing, and sorting control logic.
- Built an ESP32/C++ mobile robot with autonomous navigation, sensor-driven object detection, mechanical claw actuation, and Git-based version control.

Additional Experience

Sailing Instructor & Race Team

June 2023 – Aug 2024

Royal Canadian Yacht Club / OHSF — Instructed youth sailing; raced 29ers and C420s nationally.

Violinist & Volunteer Instructor

2012 – 2024

Royal Conservatory of Music / The Hammer Band — RCM Grade 9; performed at Koerner Hall; mentored junior musicians.

Skills

Technical: Java, C/C++, Python, PyTorch, MATLAB, Git, SolidWorks (CSWA), Onshape, Excel, Datadog

Languages: Spanish (fluent), French (intermediate)

Interests: Racquet sports (tennis, squash, and pickleball), running, travel.